Find and Fix: Legs, Hypotenuses, and the Pythagorean Theorem
Answer Key

Grade 8 / Pre-Algebra

Quick Answers

1. 15 cm
2. 24 cm
3. the corrected length = 12 cm, —
4. 10
5. the corrected height = 15 ft, —
6. both solutions of the equation = −12, 12, —
7. 26 m
8. the correct length of the second leg = 21 cm, —

What is verified

4 of 8 problems fully machine-checked against SymPy.
— marks an answer only you can judge — problems 3, 5, 6, 8. The worked solution states what a correct response must contain.

Curriculum

Level: Grade 8 / Pre-Algebra
8.G.B.7 – *problems 1, 2, 3, 5, 6, 7, 8*
8.G.B.8 – *problem 4*
Difficulty 1–5 of 5 across 12 tagged checks.

Common wrong answers

1. *If they got 21: added the two legs instead of adding their squares*
2. *If they got 25.96: treated the hypotenuse as a leg and added the squares instead of subtracting*
3. *If they got 13.93: substituted the hypotenuse 13 into the leg slot and added the squares*
4. *If they got 5.29: subtracted the squares of the horizontal and vertical gaps instead of adding them*
5. *If they got 18.79: added the squares although the ladder is the hypotenuse, not a leg*
7. *If they got 21.82: subtracted the squares, treating the diagonal as if it were one of the sides*
8. *If they got 35.23: added the squares, putting the hypotenuse 29 into a leg slot*

1. Right triangle with legs 9 cm and 12 cm. Find the hypotenuse.

Solution: Both given sides are legs, so the hypotenuse is the side we are adding towards: $c = \sqrt{9^2 + 12^2} = \sqrt{81 + 144} = \sqrt{225}$. $\boxed{15 \text{ cm}}$

2. Right triangle with hypotenuse 25 cm and one leg 7 cm. Find the other leg.

Solution: Here the longest side is given, so the missing side is a leg and the squares are subtracted: $b = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576}$. $\boxed{24 \text{ cm}}$

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3. Jamal wrote $\sqrt{13^2 + 5^2} = 13.93$ cm for the missing side of a right triangle with hypotenuse 13 cm and leg 5 cm.

(a) Solution: The 13 is the hypotenuse, so it sits alone on the large side of $a^2+b^2 = c^2$: $\sqrt{13^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144}$. $\boxed{12 \text{ cm}}$

(b) Diagnosis (instructor-judged). A correct response says that Jamal treated the hypotenuse as a leg: he added the two squares when finding a leg calls for subtraction. Accept any wording that names 13 as the hypotenuse and the operation as subtraction. Note the size check: his answer 13.93 is longer than the hypotenuse itself, which is impossible. The corrected value alone is not a diagnosis.

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4. Find the distance between $(-2, 1)$ and $(4, 9)$.

Solution: The horizontal gap is $4 - (-2) = 6$ and the vertical gap is $9 - 1 = 8$. Those two gaps are the legs of a right triangle and the distance is its hypotenuse, so the squares are added: $d = \sqrt{6^2 + 8^2} = \sqrt{36 + 64} = \sqrt{100}$. $\boxed{d = 10}$

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5. Priya wrote $\sqrt{17^2 + 8^2} = 18.79$ ft for the height a 17 ft ladder reaches with its foot 8 ft from the wall.

(a) Solution: The ladder is the slanted side, so it is the hypotenuse; the wall height is a leg: $\sqrt{17^2 - 8^2} = \sqrt{289 - 64} = \sqrt{225}$. $\boxed{15 \text{ ft}}$

(b) Diagnosis (instructor-judged). A correct response says Priya added the squares as though the ladder were a leg. The ladder is the longest side of the triangle, so it is the hypotenuse, and finding a leg means subtracting. Her answer of 18.79 ft would put the top of the wall further away than the ladder is long. Accept any wording that identifies the ladder as the hypotenuse and the operation as subtraction.

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6. Solve $x^2 + 81 = 225$ and say which solution can be the leg.

Solution: Subtract to isolate the square, then take both square roots:

$$x^2 = 225 - 81 = 144, \qquad x = \pm 12$$

$$\boxed{x = -12 \quad \text{or} \quad x = 12}$$

Reason (instructor-judged). A correct response chooses 12 and explains that squaring destroys the sign, so the equation has two solutions, but a side of a triangle is a length and cannot be negative. Accept any wording that rejects -12 because lengths are positive; a circled answer with no reason earns no credit.

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7. A garden 24 m by 10 m: how long is the corner-to-corner path?

Solution: The path is the diagonal of the rectangle, and the two sides meet at a right angle, so the sides are the legs and the path is the hypotenuse: $\sqrt{24^2 + 10^2} = \sqrt{576 + 100} = \sqrt{676}$. $\boxed{26\text{ m}}$

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8. Leg 20 cm, hypotenuse 29 cm. Marisa wrote 35.23 cm; Ben wrote 21 cm. Who is right, and how could you tell?

(a) Solution: The missing side is a leg, so subtract: $\sqrt{29^2 - 20^2} = \sqrt{841 - 400} = \sqrt{441}$. Ben is right. $\boxed{21\text{ cm}}$

(b) Reason (instructor-judged). A correct response gives the size argument: the hypotenuse is the longest side of a right triangle, so every leg must be shorter than 29 cm. Marisa's 35.23 cm is longer than the hypotenuse, so it can be rejected before any arithmetic is rechecked. Accept any wording that compares the answer with the hypotenuse; naming Ben without a reason earns no credit.